

RESEARCH ARTICLE

Modeling Energy Communities: A Case Study of Quantum Approximate Optimization on a Superconducting Processor

MATEO ALONSO^{ID}, GUILLERMO RUBIÑOS RODRÍGUEZ, PABLO DÍEZ-VALLE, ANA GARBAYO^{ID}, XELA GARCÍA-SANTIAGO^{ID}, AND GONZALO BLÁZQUEZ GIL^{ID}

Instituto Tecnológico de Galicia, 15003 A Coruña, Spain

Corresponding author: Gonzalo Blázquez Gil (gblazquez@itg.es)

This work was supported in part by the CERVERA Research Program of the Industrial and Technological Development Centre (CDTI) of Spain, through the Research Project Alianza de Centros Tecnológicos para la innovación en aplicaciones Quantum Computing para la empresa (ARQA) under Grant CER-202331031.

ABSTRACT This work explores the use of variational quantum algorithms to optimize energy distribution among users in energy communities, using real data from a community lab. This requires integrating various energy sources, storage solutions, and the ability to respond to variations in demand within energy systems, while maintaining the capacity to adapt to the variability of renewable energy sources. Given the increasing complexity as the problem size grows and the limitations of classical computing methods in addressing it, we study an approach that models the problem using QUBO and solves it applying quantum optimization algorithms. Our analysis includes a comparative study of two variational techniques, QAOA and VQE, and the implementation of the former on Qmio, an actual superconducting quantum processor. Although deploying QAOA on Qmio required specific adaptations to the hardware, we were able to achieve a shift of the probability distribution towards lower-energy solutions without error mitigation techniques, which highlights both QAOA's potential and the intrinsic challenges of quantum optimization.

INDEX TERMS Energy communities, energy efficiency, optimization models, quantum algorithms, quantum computing, quantum optimization.

I. INTRODUCTION

The European Green Deal has outlined several objectives for 2030, including enhancing energy efficiency, and expanding the share of energy produced from renewable sources [1]. In order to achieve these goals, future energy systems will need to seamlessly combine different energy sources, distributed generation methods, storage solutions and user participation through demand response programs in an efficient and adaptable manner. This adaptability is key, due to the increasing reliance on renewable energy, especially from variable sources like wind and solar, that poses challenges for maintaining a stable energy supply. One potential approach to solve these issues is the concept of energy communities,

groups of users that come together to produce, consume, store and manage their own renewable energy [2], [3], [4].

As energy communities expand, the quest for an optimal energy distribution that meets the community's needs in a cost-effective and efficient manner becomes an increasingly complex problem [5]. One approach for addressing this optimization problem is to formulate it as a Quadratic Unconstrained Binary Optimization (QUBO) where decision variables are represented in binary form [6]. However, finding the optimal solution for a QUBO problem using algorithms that search over all possible configurations to obtain the best result (like branch-and-bound, dynamic programming, and integer linear programming) can require a vast amount of computational time owing to the increasing problem size, making it an NP-hard problem. Under these circumstances, it is necessary to turn to new optimization methods capable of

The associate editor coordinating the review of this manuscript and approving it for publication was Qiang Li^{ID}.

addressing dynamic factors, such as variable energy sources, fluctuations in demand, and the need for real-time decision-making [7]. In these cases, approximate or heuristic methods are preferred, since their aim is to find near-optimal solutions in a reasonable time frame [7], [8], [9], [10].

In this context, quantum computing's potential to enhance combinatorial optimization could play a transformative role in energy management systems, especially in scenarios requiring rapid adjustments and real-time adaptability [11], [12], [13], [14]. As an example of this potential, Grover's quantum algorithm for unstructured search achieves a square-root speed-up over classical algorithms, although it requires fault-tolerant quantum computing for its implementation [15]. Nevertheless, potential advantages using current noisy quantum computers have to be analyzed for each particular case [16]. This fact suggests that a quantum approach to the problem may offer advantages in terms of speed and efficiency compared to classical computing methods [17], [18]. One option to address these types of problems is to employ quantum approximate optimization algorithms (QAOA), which utilize both classical and quantum elements to obtain solutions [19], [20]. Their noise tolerance makes variational algorithms like QAOA particularly suitable for near-term applications on Noisy Intermediate-Scale Quantum (NISQ) processors [21].

The plan of this paper is organized as follows. In Section II, we shall provide some theoretical background for the methods used later on. In particular, we define the QUBO formulation and present two quantum variational algorithms, VQE and QAOA, along with some common metrics to evaluate their performance. Section III describes the optimization problem, comparing different approaches such as the definition of battery energy at each moment versus the energy during the charging or discharging process. The aim is to obtain the optimal times for charging or discharging batteries, which represent energy storage, and operating flexible loads, energy needs that can be met within a defined time period and can be interrupted, considering the predictions for renewable generation and energy demanded by the users. Then, the cost function is transformed into an Ising Hamiltonian that is solved using Variational Quantum Algorithms. We complete the definition of the problem with some insights on the dataset, real data from an energy community lab. Section IV is devoted to a detailed analysis of our results, including a comparative analysis of QAOA and VQE, metrics used for evaluation and the implementation on an actual quantum processor. Finally, in Section V we summarize our main results and discuss possible extensions of our work.

II. METHODS

In this section, we will present the mathematical framework for modeling the energy community, as well as the variational algorithms we considered in our approach.

A. QUADRATIC UNCONSTRAINED BINARY OPTIMIZATION FORMULATION

In order to solve optimization problems using variational quantum algorithms, the cost function of the problem needs to be codified into a Hamiltonian. Finding the ground state of the Hamiltonian is thus equivalent to finding the minimum of the objective function. A common approach to obtain a Hamiltonian is to define the problem as a QUBO, a mathematical model that uses binary variables, includes no explicit constraints and has a quadratic cost function. This means that the cost function incorporates linear terms, pairwise interactions between variables and potentially constant terms. The standard form of a QUBO objective function is,

$$f(x) = \sum_{i \leq j} Q_{ij} x_i x_j, \quad (1)$$

where $x_i \in \mathbb{B}$ are binary variables and $Q_{ij} \in \mathbb{R}$ are linear (quadratic) coefficients when $i = j$ ($i \neq j$). The QUBO cost function inherently lacks explicit constraints. However, they can still be integrated when reformatted as penalty terms within the function [6]. This penalty terms evaluate to zero when satisfied but become positive when violated. Equality constraints of the type $fg(x) = a$, where $g(x)$ represents a function of binary variables, x , are reformulated into cost terms as,

$$(g(x) - a)^2 = 0. \quad (2)$$

The approach shown in (2) implements the equality constraint while avoiding the inclusion of extra binary variables. Inequality constraints, however, do require additional binary variables, known as slack variables. An inequality constraint $h(x) \leq b$, can be incorporated into the QUBO formulation as a penalty term of the form,

$$(h(x) + S(s) - b)^2 = 0, \quad (3)$$

where $S(s)$ is defined in terms of the slack variables s that must be optimized to minimize this penalty. Incorporating (3) into the QUBO cost function effectively satisfies the inequality constraint. Additionally, each penalty term is multiplied by a parameter, $\lambda \in \mathbb{R}$, which balances constraint satisfaction and objective function minimization. Without appropriate scaling, the optimizer might minimize the objective without fully satisfying these constraints or, conversely, ignore the objective and only focus on finding feasible solutions.

This form of representation can apply to a wide range of optimization problems. In the context of quantum computing, QUBO problems are frequently utilized due to their essential connection to the Ising model [22]. In physics, the Ising model is used to describe the Hamiltonian function associated with atomic arrangements in a lattice. The classical description of this function is,

$$H(z) = - \sum_{i < j} J_{ij} z_i z_j - \sum_{i=1}^N h_i z_i, \quad (4)$$

with $z_i = \pm 1$ being spin variables and J_{ij} and h_i real coefficients, equivalent to the quadratic and linear terms respectively of the QUBO function. A classical spin system has only two spin states, ± 1 , whereas in a quantum Ising model the spin becomes an operator with eigenvalues ± 1 . Thus, in order to promote (4) to a quantum Hamiltonian, let us substitute the spin variables with Pauli $\hat{Z}$ matrices,

$$\hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (5)$$

The quantum Ising Hamiltonian is thus,

$$H(\hat{Z}) = - \sum_{i < j} J_{ij} \hat{Z}_i \otimes \hat{Z}_j - \sum_{i=1}^N h_i \hat{Z}_i. \quad (6)$$

Thus, the steps to go from the QUBO cost function (1) to the Hamiltonian (6) involve first changing the binary variables into spin variables, using $z = 1 - 2x$, and then promoting these variables to Pauli $\hat{Z}$ operators as described above. The resulting Hamiltonian is an observable that encodes the QUBO problem on its diagonal. After this transformation, finding the lowest energy or ground state corresponds to solving the QUBO optimally. It can be shown that the coefficients in (6) relate to the ones in (1) as

$$J_{ij} = -\frac{1}{4}Q_{ij}, \quad (7)$$

$$h_i = \frac{1}{2} \left(Q_{ii} + \sum_{j>i} Q_{ij} \right), \quad (8)$$

plus a constant, offset term. Numerous quantum optimization techniques capitalize on this relationship, often necessitating a transformation of the QUBO problem into an Ising Hamiltonian.

B. QUANTUM APPROXIMATE OPTIMIZATION

Quantum computing is a multidisciplinary field that aims to solve complex computational tasks using quantum mechanical systems. Quantum computers use the quantum bit, or *qubit*, as the basic unit of information, which makes it possible to exploit quantum phenomena such as superposition and entanglement to process information in a fundamentally different way from classical computers.

The execution of quantum algorithms such as Grover's search algorithm is likely to require fault-tolerant technology capable of working correctly even in the presence of errors [15]. Nevertheless, the development of quantum computers is at an early stage and one can encounter multiple sources of error. As a consequence, the absence of effective correction mechanisms prevents the implementation of deep and complex quantum circuits.

However, quantum computers may still offer advantages in certain applications in the so-called Noise Intermediate Scale Quantum (NISQ) era, before fault-tolerant quantum computers become available [21]. The most popular approach to achieve such a milestone is the Variational Quantum

Algorithms (VQAs) [23], [24], [25]. These algorithms leverage a hybrid quantum-classical approach to reduce the complexity of quantum circuits at the cost of including a classical optimization procedure. The framework of the Variational Quantum Algorithms provides great versatility, while they share the same common scheme (see Fig. 1).

The quantum computer prepares and characterizes a parameterized quantum circuit or *ansatz*,

$$|\Psi(\theta)\rangle = \prod_{l=1}^p U(\theta_l) |\Psi_0\rangle, \quad (9)$$

where U are sequence of unitary quantum operations depending on the variational angles θ_l , $|\Psi_0\rangle$ is some initial quantum state, and p is the number of layers of the algorithm. The variational angles θ are tuned using a classical optimization method and an objective function $C(\theta)$ that encodes the practical problem to solve. A successful run of the algorithm returns optimal angles θ_{opt} and a quantum circuit approximating the solution of the problem, $|\Psi(\theta_{opt})\rangle \approx |\Psi_{sol}\rangle$.

The success of Variational Quantum Algorithms lies in the efficient classical optimization of the circuit parameters. Several references have studied the suitability of different optimization methods. However, the most optimal choice is still an open research question [25]. Optimizations using only gradient-based methods, such as L-BFGS [26], have been shown to generally fail in realistic scenarios when gradients must be estimated from a finite number of quantum processor measurements [27], [28], [29]. Strategies such as the parameter shift rules and quantum natural gradients provide interesting alternatives [30], [31]. Gradient-free optimizers have also been widely used to optimize variational quantum circuit parameters [32]. Based on their optimization performance and the number of quantum circuit evaluations required, the experiments in the manuscript are performed using the gradient-free optimizers COBYLA [33] and Differential Evolution [34] for numerical simulations and experimental QMIO results respectively.

The Variational Quantum Eigensolver (VQE) is a Variational Quantum Algorithm introduced as a method to approximate the ground state for a given Hamiltonian [35]. Although VQE was originally proposed to address problems in quantum chemistry, it has also been studied in the context of optimization [28], [36]. In this algorithm, the *ansatz* is not defined by the problem at hand and can be adapted to constraints such as hardware connectivity. The specific single- and two-qubit gates that compose the *ansatz* define the expressibility of the quantum circuits and the parameter trainability, thus highly affecting the performance of the algorithm. The variational angle optimization is performed by minimizing the expectation value of the Hamiltonian $\hat{H}_p$ that encodes the problem,

$$C(\theta) = \langle \Psi(\theta) | \hat{H}_p | \Psi(\theta) \rangle. \quad (10)$$

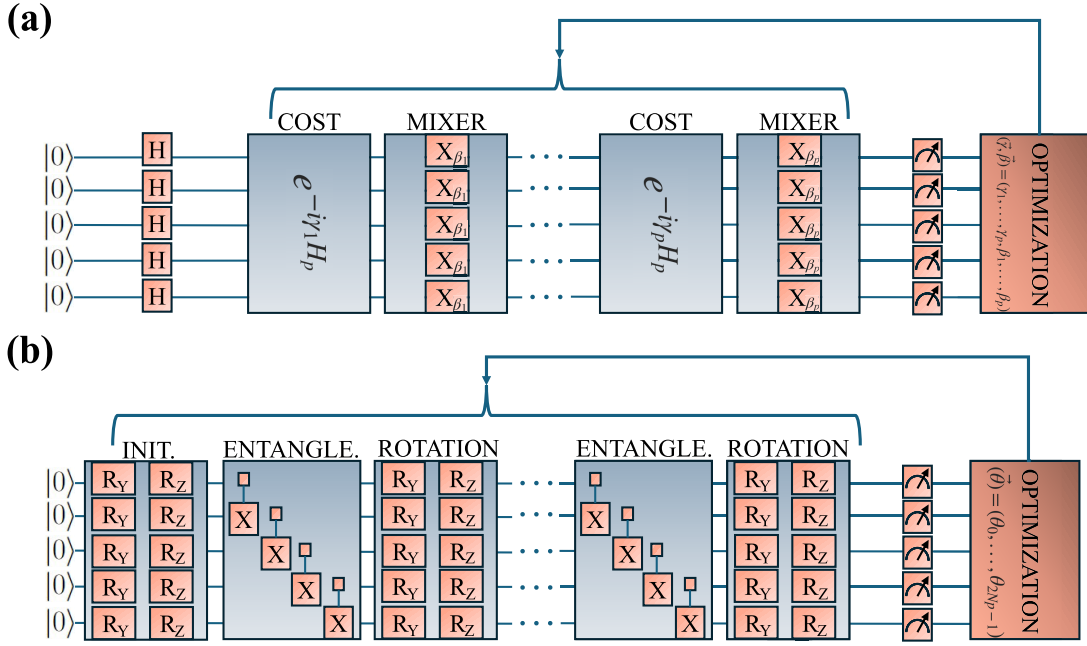


FIGURE 1. (a): QAOA ansatz, which starts with the application of Hadamard gates (H in the figure), setting the initial state for the optimization process. The *COST* layer applies rotations based on the specific Hamiltonian defined for the problem. This step is followed by a *MIXER* layer, used to explore different solution states that are controlled by the parameter β_j . The number of applications of *COST* and *MIXED* layer is determined by the depth of the algorithm. Finally, a classical optimization loop is used to find the parameters that minimize the cost function. (b): VQE ansatz, which starts by applying the parameterized gates R_Y and R_Z to the N qubits. Afterward, a set of p repetitions of the *ENTANGLEMENT* and *ROTATION* layers are applied. The *ENTANGLEMENT* layer correlates all qubits linearly using *CNOT* gates while the *INITIALIZATION* and the *ROTATION* layers search for the minimized cost function controlled by the parameters θ .

However, in the field of combinatorial optimization, the Quantum Approximate Optimization Algorithm (QAOA) stands out as the most investigated variational technique. It was proposed by Farhi et al. [19] to tackle combinatorial optimization problems, although it has also been shown to have interesting properties from the generative perspective [37], [38], [39]. Given an optimization problem whose solution is encoded in the ground state of an Ising Hamiltonian $\hat{H}_p$ as explained in Sec. II-A, the QAOA employs a specific structured ansatz inspired by a trotterized adiabatic quantum computing evolution (see Fig. 1),

$$|\Psi_{\text{QAOA}}(\boldsymbol{\gamma}, \boldsymbol{\beta})\rangle = \prod_{l=1}^p \left(e^{-i\beta_l \hat{H}_x} e^{-i\gamma_l \hat{H}_p} \right) H_d^{\otimes N} |0\rangle^{\otimes N}, \quad (11)$$

where $\boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_p)$ and $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)$ are the variational parameters, H_d is the Hadamard gate, and $\hat{H}_x$ is a non-diagonal Hermitian operator typically called the mixing Hamiltonian. The most common choice of the mixing Hamiltonian is the sum of single-qubit Pauli $\hat{X}$ operators, $\hat{H}_x = \sum_{j=1}^N \hat{X}_j$. The variational angles are then adjusted to approximate the solution of the optimization problem by the minimization of the following cost function,

$$C(\boldsymbol{\gamma}, \boldsymbol{\beta}) = \langle \hat{H}_p \rangle = \langle \Psi_{\text{QAOA}}(\boldsymbol{\gamma}, \boldsymbol{\beta}) | \hat{H}_p | \Psi_{\text{QAOA}}(\boldsymbol{\gamma}, \boldsymbol{\beta}) \rangle, \quad (12)$$

which is the expectation value of the problem Hamiltonian. As shown in Section IV, the number of layers p largely determines the accuracy of the algorithm.

Although the Quantum Approximate Optimization Algorithm has yet to demonstrate a practical advantage over classical optimization algorithms, it has gathered attention in recent years due to some remarkable results. In terms of scalability, the number of variational parameters of QAOA is independent of the size of the problem, which depends solely on the number of layers p needed to achieve the desired accuracy. Furthermore, for models such as the Sherrington-Kirkpatrick, studies have shown a concentration in the values of the objective function across different problem instances with fixed QAOA parameters [40], [41], [42]. This suggests that optimized parameters from one instance may be applicable to other instances. In terms of performance, QAOA has been able to outperform Quantum Annealing in certain cases even at depth $p = 1$ [43], and it has been shown to overcome the limit of the Goemans-Williamson approximation ratio in maximum cut problems for a larger number of p [44].

In addition to theoretical results, the QAOA has also been demonstrated experimentally on well-defined standard problems [45], [46], [47]. Although these experiments are essential for the understanding of the algorithm's performance, they lack a direct translation to the industrial sector. In this manuscript, we address this challenge and

provide experimental results of the QAOA on a practical problem of great interest in the energy industry. Other research has also tested the applicability of QAOA in extractive summarization [48], portfolio optimization [49], or manufacturing [50]. Our work aims to provide new tools and extend this knowledge to the optimization of energy community operations.

C. METRICS

The results of the VQAs are a distribution of energies obtained by sampling the ansatz with the optimal parameters found. The final output of the algorithm is the solution from the distribution which yields the lowest energy. Nevertheless, using this best solution as an indicator of the result's quality is usually misleading due to its numerical instability, especially when dealing with smaller sample sizes. Thus, to evaluate the performance of the algorithm, the complete distribution of the energy values is often considered, along with the classically computed optimal solution, $E_{\min}$. This is obtained by iterating over all possible solutions and calculating the energy for each one. For M number of samples, the energies obtained for each solution x_k with $k = 1, 2, \dots, M$ are ordered as $E_1(x_1) \leq E_2(x_2) \leq \dots \leq E_M(x_M)$. For simplicity, the considered sampled energies and the optimal solution are all considered to be negative values. The Best Measurement Ratio (BMR), used to assess the similarity between the best solution sampled compared to the known optimum, is defined as,

$$\text{BMR} = \frac{|E_1(x_1)|}{|E_{\min}|}, \quad (13)$$

where a value different from 1 indicates that the best quantum solution is not the actual optimum. Another useful metric often used to evaluate the results of optimization problems is the Approximation Ratio (AR) [51]. It is defined as the expectation value of the Hamiltonian for the state solution normalized by the minimum energy to the range $[0, 1]$,

$$\text{AR} = \frac{|\langle \Psi(\theta^*) | \hat{H}_p | \Psi(\theta^*) \rangle|}{|E_{\min}|}, \quad (14)$$

where $\hat{H}_p$ is the Hamiltonian that encodes the optimization problem and $\Psi(\theta)$ is the *ansatz* parameterized by the vector θ^* resulting from the optimization. The expectation value can be approximated as the mean of the sampled energy distribution. Hence,

$$\text{AR} \simeq \frac{\sum_{k=1}^M |E_k|}{M|E_{\min}|}. \quad (15)$$

As this metric considers the complete set of samples, it represents the average quality of the optimization results. Two other metrics are defined that focus on a section of energies at the lower end, rather than on the whole distribution. These are the Conditional Value at Risk Ratio

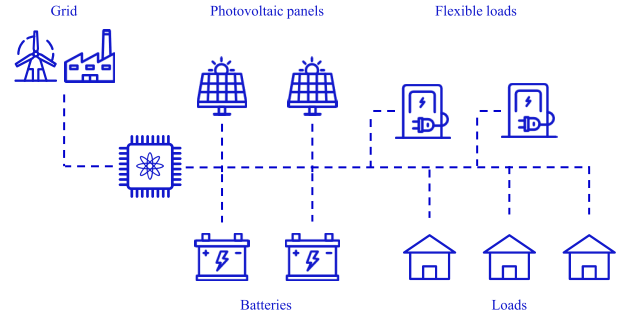


FIGURE 2. Scheme of the energy community including the photovoltaic panels used for renewable energy generation, batteries for energy storage, flexible loads for optimizing energy usage and programmable electronic loads to generate consumption in specific intervals. The whole system is connected to the electrical grid, which allows one to sell or buy energy according to the needs of the community.

(CVaRR) [52] and the Gibbs objective function Ratio (GR) [53]. The CVAR ratio is defined as

$$\text{CVaRR} = \frac{\sum_{k=1}^{\lceil \alpha M \rceil} |E_k|}{\lceil \alpha M \rceil |E_{\min}|} \quad (16)$$

where α is a parameter that determines the fraction of the distribution that is selected, such that when $\alpha = 0$ then $\text{CVaRR} = \text{BMR}$ and if $\alpha = 1$ then $\text{CVaRR} = \text{AR}$. The Gibbs ratio is defined as

$$\text{GR} = \frac{\ln \langle \Psi(\theta^*) | e^{-\eta \hat{H}_p} | \Psi(\theta^*) \rangle}{\eta |E_{\min}|} \quad (17)$$

with $\eta > 0$ being a tunable parameter that modifies the value of the Gibbs function, $f_G = \ln \langle \Psi(\theta^*) | e^{-\eta \hat{H}_p} | \Psi(\theta^*) \rangle$, to give more or less relevance to the lower energy states. In this case, when $\eta \rightarrow 0$ then the Gibbs function approximates to the expected value of the Hamiltonian, and thus $\text{GR} \rightarrow \text{AR}$ and when $\eta \rightarrow \infty$ then the Gibbs function approximates the ground state of the Hamiltonian and thus $\text{GR} \rightarrow \text{BMR}$. Both of these metrics, CVaR and Gibbs ratio, give a result between the BMR and the AR.

III. PROBLEM DEFINITION

This paper is devoted to the definition of the energy cost of running an energy community that includes renewable energy sources, energy demand forecasting, a battery for storing surplus energy, and flexible loads. A scheme of this energy community is depicted in Fig. 2. These flexible loads can be scheduled within specific time intervals, do not require continuous operation, and can be paused when necessary. The battery allows excess generated energy to be saved for use when demand exceeds generation.

In this section, we model this system as a QUBO using binary decision variables. We will then transform it to an Ising Hamiltonian that can be solved using quantum algorithms, and present the dataset.

A. MATHEMATICAL MODEL

The optimization of the components of the energy community focuses on the batteries, to decide when to charge or discharge them, and the flexible loads, to determine the appropriate times to connect them. Since the problem is modeled as a combinatorial optimization, the time dimension is discretized into N intervals with a fixed duration of Δt .

In order to model the flexible loads, the binary variables x_i^l are defined as the state of load l at a time step i , with $l = 1, 2, \dots, L$ and $i = 0, 1, \dots, N$. Considering that the energy demanded by each load for the duration of a time interval is E_l , the energy cost of all flexible loads at each time step is, as defined in [12], [17],

$$E_i^F = \sum_{l=0}^L E_l x_i^l. \quad (18)$$

We include the following constraint to ensure that each load is switched on for the required amount of time,

$$\left(\sum_{i=1}^N x_i^l - \delta_l \right)^2 \quad \forall l, \quad (19)$$

where δ_l are the number of time intervals that are needed to satisfy the demand of load l [12], [17].

For battery modeling, several approaches are available. One method involves representing the total energy stored at each time step, as is done in [13] using continuous variables. To replicate this strategy using binary variables, the capacity of the battery at each time step is discretized into $D \in \mathbb{Z}^+$ capacity states separated by an interval of size $\Delta c = 1/(D-1)$, and the fraction of the battery capacity, $c \in [0, 1]$, is determined using the binary variables x_{ij}^b , where $j = 0, \dots, D-1$, by assigning each interval, $j\Delta c$, to a variable at each time step,

$$c_i = \Delta c \sum_{j=0}^{D-1} j x_{ij}^b \quad \forall i. \quad (20)$$

If at a given time, i , the battery has a capacity of $j\Delta c$, then the binary variable x_{ij} is equal to 1 and the remaining x_{ik} with $k \neq j$ must be 0. This condition means that at each time step the battery can only be at one particular state of charge. To ensure this is true, the following penalty term has to be included in the cost function,

$$\sum_{i=1}^N \left(\sum_{j=0}^{D-1} x_{ij} - 1 \right)^2 \quad \forall i. \quad (21)$$

Using this definition of the battery capacity, the energy with which the battery is charged (if positive) or discharged (if negative) is,

$$E_i^{\text{bat}} = E_{\text{max}}^{\text{bat}} \Delta c \sum_{j=0}^{D-1} j (x_{ij}^b - x_{i-1j}^b) \quad \forall i, \quad (22)$$

where the value c_0 is an initial condition that must be set. To ensure that this energy does not exceed the maximum

energy than can be stored or extracted from the battery during a one hour interval, another constraint is necessary, $-E^{\text{lim}} \leq E_i^{\text{bat}} \leq +E^{\text{lim}}$, where E^{lim} is the energy limit, which is considered to be the same for charge and discharge for simplicity and because that is the case for the real batteries considered in the scenarios. The resulting penalty term that needs to be included is,

$$\sum_{i=1}^N \left(E_i^{\text{bat}} - E^{\text{lim}} + E_{\text{max}}^{\text{bat}} \Delta c S_i^E \right)^2, \quad (23)$$

where the slack term, $S_i^{c,d}$, is

$$S_i^E = \sum_{k=1}^{N_s^E-1} 2^{k-1} s_{ik}^E + \left(V - 2^{N_s^E-1} \right) s_{iN_s^E}^E, \quad (24)$$

where $N_s^E = \lceil \log_2(V) \rceil \in \mathbb{Z}^+$ is the number of binary slack variables, s_i^E , that are needed to compensate for the $V = 2E^{\text{lim}}/E_{\text{max}}^{\text{bat}} \Delta c + 1$ feasible integer values at a given time, both for the minimum and the maximum values. Finally, the QUBO problem reads,

$$\begin{aligned} \min_{x,y,s} & \left[\sum_{i=1}^N (p_i^+ E_i y_i + p_i^- E_i (1 - y_i)) \right. \\ & + \lambda_F \left(\sum_{i=1}^N x_i^l - \delta_l \right)^2 \\ & + \lambda_s \sum_{i=1}^N \left(\sum_{j=0}^{D-1} x_{ij} - 1 \right)^2 \\ & + \lambda_E \sum_{i=1}^N \left(E_i^{\text{bat}} - E^{\text{lim}} + E_{\text{max}}^{\text{bat}} \Delta c S_i^E \right)^2 \\ & \left. + \lambda_y \left(\sum_{i=1}^N \epsilon (2y_i - 1) - E_i \right)^2 \right], \quad (25) \end{aligned}$$

with $\lambda_F, \lambda_E, \lambda_y \in \mathbb{R}^+$ being the parameters that can adjust the weight of each penalty term in the cost function.

Another possibility, which reduces the number of necessary variables while still allowing for a lower value of Δc , is to directly model the change in capacity. Here, we will introduce two binary variables: x_i^c (x_i^d) is used to determine when the capacity of the battery increases (decreases) and thus the battery is charging (discharging). In the case that both variables have the same value, then the capacity remains the same from one time step to the next. The battery capacity at any given time is then,

$$c_i = c_0 + \Delta c \sum_{j=1}^i (x_j^c - x_j^d), \quad (26)$$

where Δc is the fraction that the battery capacity can increase or decrease at each time step and c_0 is the initial value for the capacity. The value of Δc is fixed and must comply with the battery specifications. Defining the capacity of the battery

as in (26) requires the inclusion of the constraint $0 \leq c_i \leq 1$ to prevent the battery from exceeding its maximum and minimum storage capacity. The penalty term added to the cost function due to this constraint is,

$$\sum_{i=1}^N \left(c_0 + \Delta c \sum_{j=1}^i (x_j^c - x_j^d) - 1 + \Delta c S_i^c \right)^2, \quad (27)$$

where S_i^c is a slack term, defined as,

$$S_i^c = \sum_{k=1}^{N_s^c-1} 2^{k-1} s_{ik}^c + \left(\frac{1}{\Delta c} + 1 - 2^{N_s^c-1} \right) s_{iN_s^c}^c, \quad (28)$$

where $N_s^c = \lceil \log_2(1/\Delta c + 1) \rceil \in \mathbb{Z}^+$ is the number of binary slack variables, s_i^c , needed to compensate for the feasible values at a given time.

The energy flow in or out of the battery is determined from the change in battery capacity from one time step to the next. Considering this and (26) yields,

$$E_i^{\text{bat}} = E_{\text{max}}^{\text{bat}} \Delta c (x_i^c - x_i^d), \quad (29)$$

where $E_{\text{max}}^{\text{bat}}$ is the energy capacity of the battery. With E_i^{bat} and E_i^F components modeled, the energy balance can be expressed as,

$$E_i = E_i^C + E_i^R + E_i^{\text{bat}} + E_i^F, \quad (30)$$

where $E_i^C \geq 0$ represents the building's energy consumption and $E_i^R \leq 0$ is the amount of renewable energy generated. The total energy cost when considering a energy buying price p_i^+ and energy surplus selling price p_i^- is,

$$J = \sum_{i=1}^N p_i^+ E_i y_i + p_i^- E_i (1 - y_i), \quad (31)$$

where the first term represents the cost of buying energy from the grid and the second term are the benefits obtained from selling excess energy. The binary variable y_i is introduced to apply the right price: when energy must be bought, the total energy E_i is positive and y_i is set to 1. Conversely, if there is surplus energy to sell, E_i is negative and y_i is set to 0. To enforce this behavior, an additional constraint on y is required in the problem formulation, such that $\text{sign}(2y_i - 1) = \text{sign}(E_i)$. Since this condition is not translatable into a quadratic expression, an approximate term has to be included instead,

$$\sum_{i=1}^N (\epsilon (2y_i - 1) - E_i)^2. \quad (32)$$

The minimum value of (32) corresponds to the desired behavior of the variable y , and the parameter ϵ is a positive real number that can be fine-tuned to improve the relevance of the variable y for large energy values. Let us note that both the cost function (31) and (32) depend on the energy, so the weight that regulates the importance of this constraint must ensure that the restriction is fulfilled while still minimizing

the energy cost. Using this second formulation, the resulting QUBO problem is

$$\begin{aligned} \min_{x,y,s} & \left[\sum_{i=1}^N (p_i^+ E_i y_i + p_i^- E_i (1 - y_i)) \right. \\ & + \lambda_F \left(\sum_{i=1}^N x_i^l - \delta_l \right)^2 \\ & + \lambda_m \sum_{i=1}^N \left(c_0 + \Delta c \sum_{j=1}^i (x_j^c - x_j^d) - 1 + S_i^c \right)^2 \\ & \left. + \lambda_y \sum_{i=1}^N (\epsilon (2y_i - 1) - E_i)^2 \right], \quad (33) \end{aligned}$$

with $\lambda_F, \lambda_m, \lambda_y \in \mathbb{R}^+$ being the parameters that can adjust the weight of each penalty term in the cost function. This optimization problem is transformed as explained in Section II-A. It must be noted that there is an extra constant term that can be discarded for the optimization, since it does not contribute to the resulting solution.

Due to the resulting size of the complete formulations, a simplified instance of the modeled problems is considered for the implementations. Firstly, the value of Δc is fixed to 0.2, to limit the charging or discharging energy to 5 kWh. The maximum number of time steps optimized are $N = 2$. Considering the scaling of both approaches seen in Fig. 3 for a low number of time steps, the second method yields a lower number of both variables and interactions. Secondly, no flexible loads are considered, as it would quickly increase the number of variables issue, particularly bearing in mind the capabilities of the quantum hardware that will be employed. Finally, only one price is considered for both buying and selling energy to the grid, so that $p_i^+ = p_i^- = p_i$. Hence, the optimization problem implemented reduces to,

$$\begin{aligned} \min_{x,y,s} & \left[\sum_{i=1}^N p_i (E_i^C + E_i^R + E_{\text{max}}^{\text{bat}} \Delta c (x_i^c - x_i^d)) \right. \\ & \left. + \lambda_m \sum_{i=1}^N \left(c_0 + \Delta c \sum_{j=1}^i (x_j^c - x_j^d) - 1 + S_i^c \right)^2 \right]. \quad (34) \end{aligned}$$

B. DATASET

The dataset used was gathered from a community energy laboratory located in the north of Spain. This laboratory includes a shared 20kW photovoltaic facility, installed on the roof of the building, and batteries with a storage capacity of 25kWh. It is also equipped with programmable electronic loads that can generate consumption within a specified time interval.

The entire system is monitored in real time, tracking energy production from the photovoltaic installation, the building's energy consumption, and the state of the batteries. Additionally, it also monitors the energy flow with the grid,

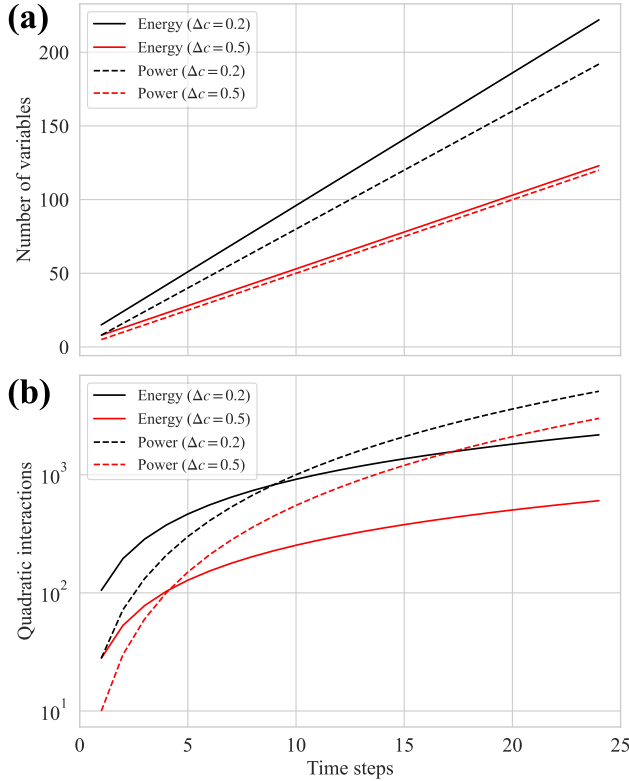


FIGURE 3. Comparison of the number of variables (a) and quadratic interactions (b) of both methods for modeling the capacity of one battery as a function of the number of time steps. The solid lines correspond to the approach 1 defined in (20) considering the constraint penalty and slack variables added by (21), while the dashed lines correspond to approach 2, defined in (26) with constraint (27). For the considered case of $\Delta c = 0.2$ and for a number of time steps lower than 9, both the number of variables and the quadratic interactions are lower for approach 2.

the electricity price, and the price for surplus energy injected into the grid.

Fig. 4 shows the variables monitored over a one-week period. The building has an average daily consumption of 75 kWh on weekdays, with higher consumption in the morning and lower values in the evening and night. The consumption drops to 20 kWh on weekends. The mean daily photovoltaic production is 25 kWh, although it varies throughout the year, with peaks in summer and lower values in winter.

From this dataset, seven different 2-hour time interval scenarios were selected for the proposed problem and are included in Table 1. These scenarios vary based on the relationship between energy consumption and photovoltaic production, the state of charge of the battery, and the electricity prices. The data from these scenarios are introduced in Eq. (34) to obtain the QUBO problem. More specifically, the energy price considered, p_i , the energy consumed, E^C , and the energy generated, E^R , by the community are all values obtained from the dataset. The sparsity of these final problems can be analyzed by calculating the density of the problem, defined as the number of existing interaction terms divided by the total number of possible interactions [54]. The

TABLE 1. Data corresponding to the seven 2-hour intervals selected from the original dataset. E^C (E^R) represents energy consumption (production) in kWh, p are the electricity prices in €/MWh and c_0 denotes charge percentage of the battery.

	Hour	E^C	E^R	p	c_0 (%)
Scenario 1	1	1.6	1.9	150	100
	2	1.5	0.6	214	
Scenario 2	1	4.2	7.9	70	100
	2	2.2	7.0	63	
Scenario 3	1	5.0	4.7	165	4
	2	6.2	7.2	149	
Scenario 4	1	2.0	0.0	110	5
	2	2.1	0.0	128	
Scenario 5	1	4.5	7.9	145	18
	2	4.3	8.4	131	
Scenario 6	1	4.1	8.4	140	72
	2	3.6	7.7	83	
Scenario 7	1	2.0	1.9	157	100
	2	1.8	0.4	196	

density for all cases with $N = 1$ is 1, while the cases with $N = 2$ is 0.666.

IV. RESULTS

Considering the data available, the value of the maximum energy capacity is set to $E_{\max}^{\text{bat}} = 25$ kWh. The Lagrange multiplier that regulates the penalty term in (34) is fixed heuristically using classical optimizations to ensure constraint satisfaction to a value of $\lambda_m = 25$. This value was selected after optimization tests across various scenarios, where $\lambda_m = 25$ performed the best among the sufficiently large multipliers that ensured the constraint is fulfilled. Additionally, several classical optimizers were tested, as in [32], choosing COBYLA for the simulated results, due to its faster optimization times while performing very similarly to other slower optimizers, and Differential Evolution (DE) [34] combined with L-BFGS-B (to polish the obtained results from the DE) for the execution on quantum hardware, since it performed significantly better than the rest. The problem was implemented for $N = 1$ and $N = 2$ time steps, resulting in a problem size of 5 and 10 variables, respectively. As in our modeling, each variable is assigned directly to a qubit. This means that each problem requires 5 and 10 qubits. These problem sizes are adjusted to fit our currently available quantum hardware constraints. For the case of the optimizations using a quantum computer simulator, the results are computed as the mean and standard deviation of the values obtained for each of the considered scenarios. When it comes to the real quantum hardware, only scenario 1 was optimized. All results take into account the intrinsic effects of shot noise [55], [56], since each evaluation of the cost function is performed by averaging 4096 samples of the quantum circuit.

1) ANSÄTZE COMPARISON

Initially, two variational techniques were considered to solve the optimization problem, namely the Quantum Approximate Optimization Algorithm (QAOA) and the Variational

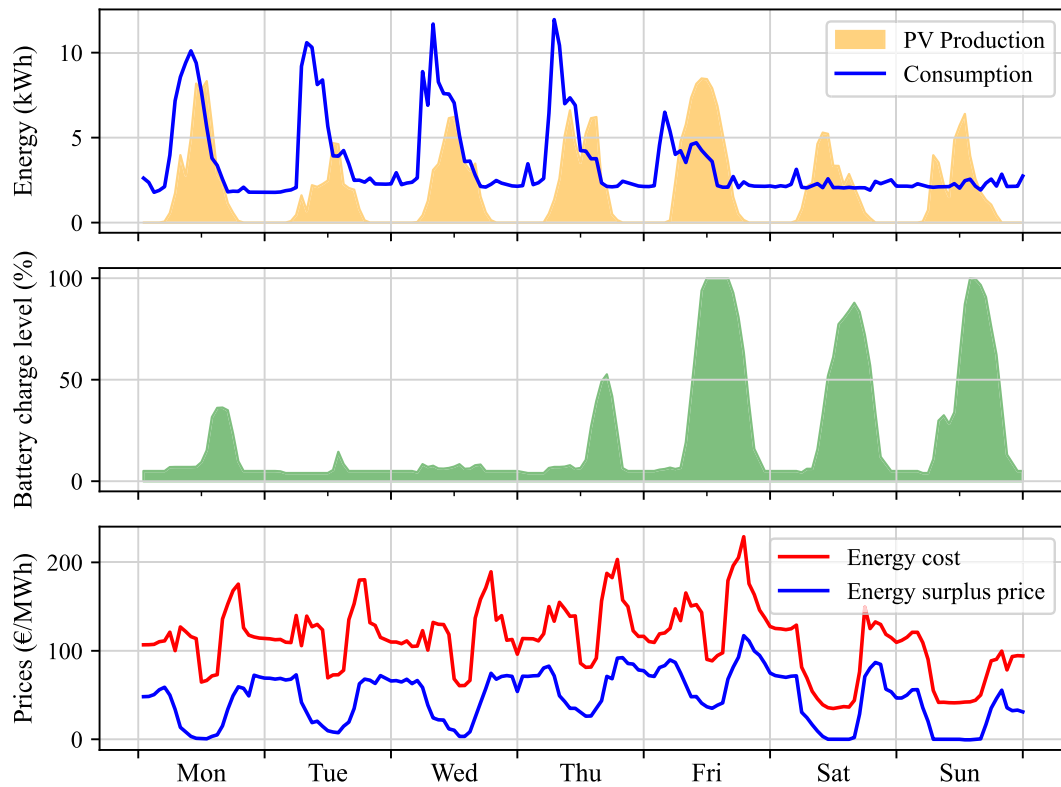


FIGURE 4. One-week period sample of the monitored variables. *Top:* photovoltaic (PV) production follows a clear daily pattern, peaking over daylight hours as expected for solar generation. We can identify a similar behavior in the varying energy consumption during the weekdays, which significantly decreases toward the weekend. *Middle:* the charge level of the batteries correlates with PV production and energy consumption, since they charge when production is high and discharge during consumption peaks. This effects can be easily seen when we compare the battery levels during the weekend (where the consumption is specially low but PV production still picks during daylight hours) with those during weekdays (when energy consumption exceeds generation). *Bottom:* both energy cost and surplus price fluctuate over the week, with some particularly high picks of the former.

Quantum Eigensolver (VQE). These algorithms differ in the quantum parametric or ansatz circuit to approximate the minimum energy state. While QAOA makes use of a problem-specific ansatz layer inspired by a trotterized adiabatic quantum computing evolution (11), we implement VQE with a hardware-efficient ansatz that adapts the entanglement structure to the available connections on the quantum computer independently of the optimization problem. In particular, the quantum circuit consists of three layers of parameterized single-qubit gates followed by two-qubit CNOTs entangling gates between nearest neighbors, as shown in Fig. 1. This is a widely used heuristic pattern for preparing trial wavefunctions for VQAs. Notice that QAOA and VQE ansätze require $2p$ and $2Np$ parameters to optimize respectively, where p is the number of layers and N is the number of qubits in the circuit.

A comparison of the metrics that have been taken into account to assess the validity of both approaches can be found in Fig. 5a. We note that while VQE achieves a slightly higher average approximation ratio (AR), the algorithm is inferior to QAOA in terms of the best minimum (BMR). This

result suggests that VQE tends to get trapped in potentially low-energy local minima, rather than reaching the ground state, more often than QAOA. In contrast, the results for QAOA support that it successfully reaches the ground state with high certainty for the small systems studied. In fact, it can be observed that the difference in the minima reached by each ansatz is also reflected in the uncertainty associated to the other metrics shown in the figure, which is notably higher for VQE than for QAOA in all the cases analyzed. In addition, the result obtained for the Conditional Value at Risk (CVaR) ratio also indicates that QAOA is able to sample a large number of lower energy states more consistently than VQE.

In Fig. 5b we corroborate the above insights by analyzing the evolution of the ground state probability amplitude in the ansatz optimization procedure, i.e., the probability of sampling the global minimum of the optimization problem from the quantum circuit. To make a fair comparison between the different instances, we round the percentage of completed iterations to ten values for QAOA and to 100 values for VQE, as they take on average 30 and 1000 iterations to converge

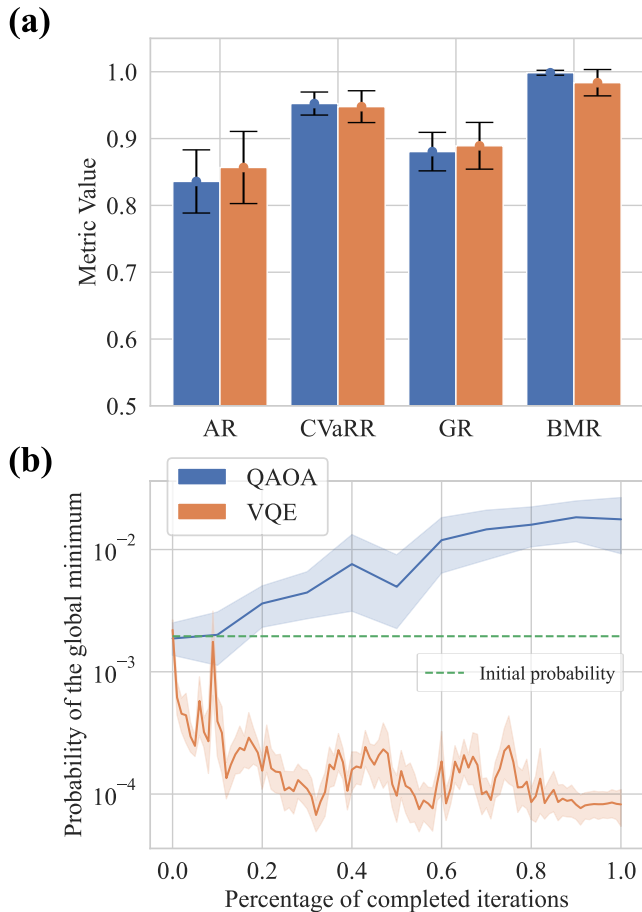


FIGURE 5. Comparison of various metrics for average results of QAOA (blue) and VQE (orange), including associated uncertainties for each metric, (a): QAOA yields better results in most of the analyzed metrics, although VQE provides a slightly higher AR. (b): QAOA consistently raises the probability of reaching the ground state, unlike VQE. The results show the standard deviation and the 95% confidence interval in the average estimation in (a) and (b) respectively.

respectively. In fact, we can observe how QAOA is able to steadily increase the ground state probability, in contrast to VQE. The explanation is that, while QAOA is less expressive than VQE and therefore only able to reach a moderate ground state amplification with one layer, VQE is easily trapped in local minima avoiding any possibility of sampling the global minima by repeating measurements of the quantum processor, in line with previous results [28].

Therefore, the use of QAOA over VQE in our experiments allows us to reduce the number of classically optimized parameters, and offers more guarantees to reach the global minimum of the optimization problem by repeated sampling. In light of these results, we conduct the remainder of the study with QAOA.

2) DEEPER ANSATZ AND NOISE EFFECT

The single-layer QAOA ansatz is not expressive enough to reach the global minimum with a moderate number of measurements for large combinatorial optimization problems,

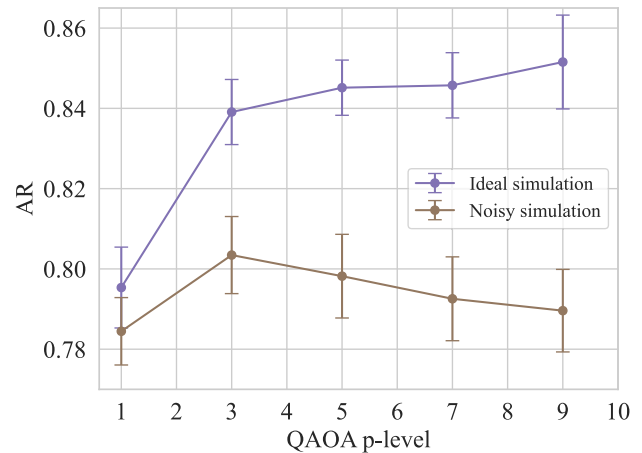


FIGURE 6. Comparison of ideal (purple) and noisy (brown) approximation ratio (AR) average results for different p-level repetitions of the QAOA layers. In the case of the ideal simulation we see a steady improvement in the AR, related to the growing ability of the ansatz to represent complex solutions. However, the performance of the noisy simulation decreases after three layers, as a result of the increment of noise effects in more complicated circuits. The error bars show the estimated standard deviation of the average.

as highlighted by existing literature [39], [57], [58]. In this regard, we analyze the improvement of QAOA performance in the energy community problem as we increase the number of ansatz layers, also taking into account possible sources of quantum hardware error.

In the first place, we run the algorithm with an increasing number of layers p using a classical simulator of an ideal quantum computer, leaving aside experimental constraints such as noise. Through these runs, we obtain the optimal variational parameters for each p and problem instance, as well as the average Approximation Ratio. Next, we run the QAOA ansatz with the optimal parameters including a simple noise model based on *Pauli errors* to see how the results can be affected by the inclusion of imperfect quantum gates. Such a model consists of an N -qubit error channel unitary created from a list of Pauli matrices and their corresponding probabilities. In particular, we implement a 4% probability of bit flip and phase flip. For 2-qubit gates we considered the tensor product of the 1-qubit Pauli errors. Both ideal and noisy simulations are implemented with the simulation tools provided by qiskit [59].

We summarize our results in Fig. 6. As expected, in the ideal scenario increasing the number of layers leads to a monotonic improvement of the Approximation Ratio derived from the growing expressibility of the ansatz. However, such a consistent enhancement does not carry over to the case where we include simple error sources. In such a scenario, we observe a decrease in performance from 3 layers onward in line with other works [17]. The performance improvement in the ideal simulation is due to the rise in the ansatz expressibility as we increase the number of layers. Nevertheless, the expressibility enhancement comes at the

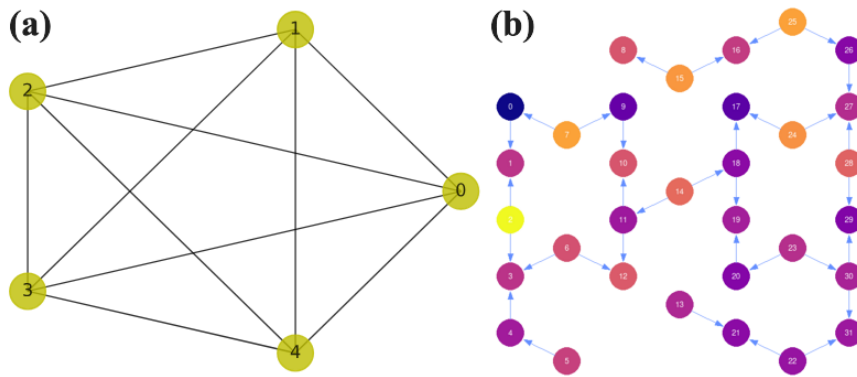


FIGURE 7. (a): Graph topology of the optimized problem corresponding to scenario 1 for one time step. The graph representation of the problem shows that it is very connected and thus very dense, as every node is connected to every other node. (b): Topology of the Qmio quantum processing unit. Currently, the best performing qubits correspond to the bottom right region.

cost of increasingly complex quantum circuits involving more gates and thus a higher susceptibility to noise effects. These results emphasize the need to find a balance between the higher expressibility of quantum circuits achieved with a larger number of layers and the accumulated errors. Low hardware error rates or error mitigation protocols seem to be needed to successfully implement QAOA at large depths.

Note that, in this subsection, we did not address other limitations of quantum hardware, such as disconnected qubits or a limited number of available quantum operations. We address this issue in the next subsection, where we optimize instances of energy community operations problem on a real quantum computer.

3) EXECUTION ON REAL HARDWARE

We now test the implementation of QAOA for energy community operations on an actual quantum device. In particular, we execute small-sized instances of the problem on Qmio, a superconducting quantum processor composed of 32 coaxm qubits (see Fig. 7). In the graph representation of the QUBO problem in Fig. 7a, where each binary variable is represented by a node in the graph and the interactions are described by weighted edges, each node is connected to each other and thus its density is equal to 1. Although the current available resources restrict the efficiency and problem complexity we can tackle, this study allows us to address in practice what the main limitations are when running real quantum approximate optimization experiments. We analyzed the result of single-layer QAOA on one step for scenario 1, which results in a 5 qubit problem.

A key aspect is the tailoring of the optimization problem to the connectivity of the quantum device. The QAOA ansatz QAOA involves mapping the connections from the Ising (6) Hamiltonian to the qubits by R_{ZZ} entangling gates,

$$R_{ZZ}(\theta) = e^{-\frac{\theta}{2}\hat{Z}\otimes\hat{Z}}. \quad (35)$$

To limit the number of additional gates, it is desirable to select the processor region that best suits the problem. However, since these are highly connected optimization problems that depart from the limited architecture of the quantum processor (see Fig. 7b), our circuit implementations require a large number of additional swap gates to allow the entanglement of disconnected qubits. With this in mind, and also trying to select the best calibrated qubits in terms of noise, we decided to map the optimization problem to qubits 19, 20, 23, 30 and 31.

Another limitation is the quantum operations that can be performed on the quantum computer, often referred to as native quantum gates. Each gate in the quantum circuit must first be decomposed in terms of these native gates by a process called *transpilation* of the circuit. The native gates of Qmio are the echoed cross-resonance (ECR), SX, R_Z and X gates. The whole process of transpilation of our quantum circuits, which involves decomposition of quantum operations and swap gates, transforms the depth of our quantum circuits from approximately 11 to about 150 depending on the transpilation pass.

We show the results obtained in Fig. 8. The classical optimizer used was DE with a population number of 15. The total number of iterations of the optimizer was 50, resulting in 1638 evaluations of the cost function, and therefore of 1638×4096 measurements of the quantum circuit. Even in the absence of error mitigation techniques, the optimization is able to slightly shift the probability distribution towards low-energy solutions, as expected.

V. DISCUSSION AND OUTLOOK

This paper analyzes the use of quantum approximate optimization to assess energy distribution optimization in energy communities, a key area for enhancing energy efficiency. The potential of quantum computing to improve optimization methods offers a promising approach to assess complex optimization problems, which traditional methods

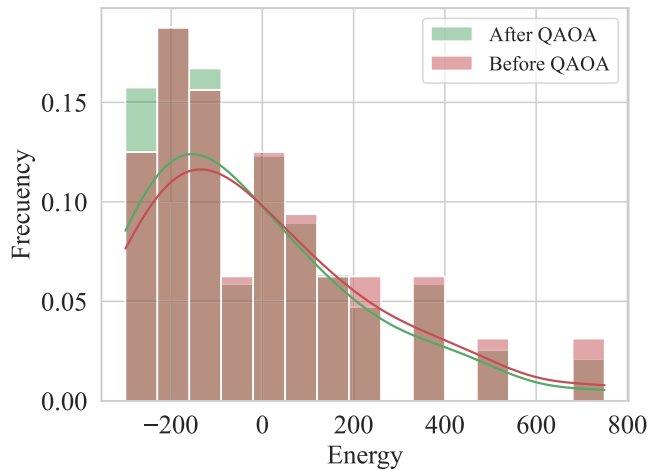


FIGURE 8. Analysis of the energy distribution before (red) and after (green) applying the QAOA ansatz on the Qmio quantum processor to a small-size energy community instance. We can see a slight amplification of the frequency of low-energy states after optimization of the single-layer QAOA, despite the absence of error mitigating techniques.

find increasingly hard to obtain exact solutions within reasonable computing power and time [16].

Firstly, in Section III we translated the optimization problem into a mathematical framework using a QUBO formulation that defined binary decision variables, which is suitable for quantum algorithms by a reformulation to an Ising Hamiltonian. We focused on modeling the flexible loads and battery use in order to optimize their operation times, taking into account energy demand and supply.

Secondly, we selected the data to test the problem definition. One should bear in mind that this study is based on real data from a community energy lab located in northern Spain. The lab's equipment includes a shared 20kW rooftop photovoltaic system and 25kWh battery storage, as well as programmable loads to simulate energy consumption. A scheme of the energy community and the optimization problem can be found in Fig. 2. Since the whole system is monitored in real time (see Fig. 4), we were able to select seven 2-hour scenarios, each reflecting different relationships between energy consumption, photovoltaic output, battery state, and energy pricing.

Thirdly, we chose the most suitable ansatz for approximating the minimum energy state. We considered two possible variational techniques, namely QAOA and VQE, differing in their ansatz circuits. In this case, we found that a single layer of QAOA outperforms VQE in most of the analyzed metrics, as can be seen in Fig. 5. In particular, we noted that the best minimum (BM) value for VQE suggests it tends to get trapped in local low-energy minima, while QAOA reliably shifts the distribution towards the lower energy states, along the line of previous works [28], [39]. This difference in minima reached by each ansatz is also evident in the greater uncertainty associated with VQE in all the metrics analyzed, although it does achieve a slightly higher average approximation ratio (AR). These insights are also supported by the evolution of

ground state probability (Fig. 5b). Considering all the above results, we focused on QAOA for the rest of the study.

However, a single QAOA layer has expressibility limitations that prevent it from reaching the global minimum with a reasonable number of measurements for general large combinatorial problems [60]. In order to explore this limitation, we reviewed the evolution of the AR when applying the algorithm with an increasing number of layers in two different scenarios: a classical simulator representing an ideal quantum environment, and another incorporating a basic noise model with Pauli errors that enabled us to probe the impact of imperfect quantum gates on our results. Our findings, depicted in Fig. 6, indicate that whereas in an ideal setting the inclusion of more layers implies an increase on the AR, the presence of noise truncates this trend, potentially due to the intensification of noise interference in more complex circuits. Hence, we find that the reliability of QAOA at greater depths will strongly depend on the attainability of very low hardware error rates or robust error-mitigation strategies.

Lastly, we apply QAOA to execute instances of the energy community problem on Qmio, an actual superconducting quantum processor with 32 qubits. The state of the art in the quantum hardware significantly reduces the problem definition one can solve in a real-life quantum computer, which means we needed to adapt to the constraints of the actual quantum device and map the problem accordingly. This step involved significant adjustments, mainly related to the selection of the qubits with best noise calibration. We found another limiting factor, related to the transpilation of the circuit, *i.e.*, the necessity of decomposing all circuit gates into the processor's native quantum gates, which resulted into an increase of the circuit gates from 11 to 150. However, as shown in Fig. 8, we were able to modify the probability distribution of a small size energy community instance to lower-energy solutions by a single QAOA layer optimization on the superconducting quantum processor, without error mitigation techniques.

A natural next step of our study is the inclusion of error mitigation techniques in the Qmio implementation. This should lead to significant improvement of the results here commented and is a necessary step if one aims to increase the size and complexity of the problem at hand. Actually, although the problem presented in Section III-A includes flexible loads and energy buying and selling prices, we have not included them in the reduced problem (34). Future advances in quantum hardware may also allow more sophisticated problems to be implemented.

From an algorithmic point of view, it would certainly be interesting to include more techniques in our energy community optimization analysis. Several variants of QAOA have been introduced in the literature, such as Multi-Angle QAOA [61], which enhances AR at the expense of increasing significantly the number of variational parameters, Recursive QAOA [62], that iteratively decreases the number of qubits in the algorithm etc. In addition, there are a number of techniques that explore the compression of the optimization

TABLE 2. Physical and measurement characteristics for selected qubits. *Meas. Freq.* stands for *Measurement Frequency*, and *Meas. Time* for *Measurement Time*. Additionally, *Fidelity* is abbreviated as *Fidel.* T1 is the relaxation time and T2 is the dephasing time.

Characteristic	Qubit				
	19	20	23	30	31
T1 (μ s)	70.46	41.84	54.31	49.55	64.93
T2 (μ s)	121.43	67.15	80.50	89.18	144.04
Meas. Freq. (GHz)	10.28	9.64	9.84	10.25	9.62
Meas. Time (μ s)	5.42	12.00	12.00	12.00	7.53
Init Fidel.	0.900	0.943	0.915	0.911	0.862
Readout Fidel.	0.930	0.905	0.880	0.876	0.900
SX Gate Fidel.	0.999	0.999	0.999	0.998	0.999
SX Duration (ns)	32.0	32.0	32.0	32.0	32.0
R_Z Fidelity	1.0	1.0	1.0	1.0	1.0

TABLE 3. Two-qubit ECR gate characteristics between selected qubit pairs. The table lists the control and target qubits involved in each gate operation, along with the corresponding fidelity and gate duration. These parameters reflect the calibration status of the hardware at the time of the experiment.

Control qubit	Target qubit	Fidelity	Duration (ns)
20	19	0.974	352
23	20	0.982	352
23	30	0.979	352
30	31	0.964	352

problem into fewer qubits, which would allow us to increase the number of time steps in our instances [63], [64], [65], [66], [67]. Another area of improvement is the definition of constraints, which here has been done by including penalty terms. Some proposals in the context of variational algorithms include the use of quantum Zeno dynamics and modified objective functions [68], [69], [70], [71], [72]. Finally, a practical advantage of QAOA in optimizing energy communities would require a comparative evaluation with classical heuristics such as Simulated Annealing. Nevertheless, any possible experimental quantum advantages, with respect to execution time, quality of the obtained solution, or sustainability, could not be achieved without the use of a larger number of qubits than those available for this work. At this point, the state of quantum hardware makes the problem sizes accessible in our experiments too small for a fair comparison.

APPENDIX

QMIO HARDWARE SPECIFICATIONS

This appendix provides detailed hardware specifications for QMIO, the quantum processor used to execute the experiments reported in this work. In particular, Table 2 summarizes single-qubit properties, while Table 3 presents two-qubit gate performance. In both tables, the qubits are numbered as in Figure 7b).

Please note that these values are subject to change over time. The data presented here reflect the hardware characteristics measured the day the experiments were performed.

ACKNOWLEDGMENT

The authors are indebted for valuable exchanges with Mariamo Mussa and Guillermo Díaz from the Galician Supercomputing Center (CESGA) Team. The authors also acknowledge CESGA for providing access to Qmio infrastructure with financing from the European Union, through the Programa Operativo Galicia 2014-2020 of ERDF_EACT EU, as part of the European Union's response to the COVID-19 pandemic.

REFERENCES

- [1] Eur. Commission. (2021). *Delivering the European Green Deal*. Accessed: Oct. 7, 2024. [Online]. Available: https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal/delivering-european-green-deal_en
- [2] (2021). *Energy Communities*. Accessed: Oct. 7, 2024. [Online]. Available: https://energy.ec.europa.eu/topics/markets-and-consumers/energy-consumers-and-prosumers/energy-communities_en
- [3] A. Caramizaru and A. Uihlein, "Energy communities: An overview of energy and social innovation," *Scientific Anal. or Rev.*, Luxembourg, Europe, Tech. Rep. KJ-NA-30083-EN-N, 2020.
- [4] N. S. Nafi, K. Ahmed, M. A. Gregory, and M. Datta, "A survey of smart grid architectures, applications, benefits and standardization," *J. Netw. Comput. Appl.*, vol. 76, pp. 23–36, Dec. 2016. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1084804516302314>
- [5] S. A. Shaikh, A. Ali, and A. D'Angola, "A review of renewable energy communities: Concepts, scope, progress, challenges, and recommendations," *Sustainability*, vol. 16, no. 5, p. 1749, Feb. 2024. [Online]. Available: <https://www.mdpi.com/2071-1050/16/5/1749>
- [6] F. Glover, G. Kochenberger, and Y. Du, "Quantum bridge analytics I: A tutorial on formulating and using QUBO models," *4OR*, vol. 17, no. 4, pp. 335–371, Dec. 2019. [Online]. Available: <https://ideas.repec.org/a/spr/annopr/v314y2022i1d10.1007s10479-022-04634-2.html>
- [7] N. Salehi, H. Martínez-García, G. Velasco-Quesada, and J. M. Guerrero, "A comprehensive review of control strategies and optimization methods for individual and community microgrids," *IEEE Access*, vol. 10, pp. 15935–15955, 2022.
- [8] Y. Zhao, "Cooperative energy optimization for multiple diversified integrated energy systems," in *Proc. 6th Int. Conf. Power Energy Appl. (ICPEA)*, Nov. 2023, pp. 253–258.
- [9] F. Lazzari, G. Mor, J. Cipriano, F. Solsona, D. Chemisana, and D. Guericke, "Optimizing planning and operation of renewable energy communities with genetic algorithms," *Appl. Energy*, vol. 338, May 2023, Art. no. 120906. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0306261923002702>
- [10] A. M. Rafi, M. Ali, M. I. Hossain, F. A. Al-Sulaiman, and M. Shafiullah, "Whale optimization algorithm for community microgrid energy scheduling," in *Proc. Saudi Arabia Smart Grid (SASG)*, Dec. 2022, pp. 1–7.
- [11] A. Ajagekar and F. You, "Quantum computing for energy systems optimization: Challenges and opportunities," *Energy*, vol. 179, pp. 76–89, May 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0360544219308254>
- [12] C. Mastroianni, L. Scarcello, and J. Settino, "Quantum computing approach for energy optimization in a prosumer community," in *Proc. IEEE Intl Conf Dependable, Autonomic Secure Comput., Intl Conf Pervasive Intell. Comput., Intl Conf Cloud Big Data Comput., Intl Conf Cyber Sci. Technol. Congr. (DASC/PiCom/CBDCom/CyberSciTech)*, Sep. 2022, pp. 1–7.
- [13] P. Schworm, X. Wu, M. Wagner, S. Ehmsen, M. Glatt, and J. C. Aurich, "Energy supply scheduling in manufacturing systems using quantum annealing," *Manuf. Lett.*, vol. 38, pp. 47–51, Sep. 2023.
- [14] G. Colucci, S. V. D. Linde, and F. Phillipson, "Power network optimization: A quantum approach," *IEEE Access*, vol. 11, pp. 98926–98938, 2023.
- [15] L. K. Grover, "Quantum mechanics helps in searching for a needle in a haystack," *Phys. Rev. Lett.*, vol. 79, no. 2, pp. 325–328, Jul. 1997, doi: [10.1103/physrevlett.79.325](https://doi.org/10.1103/physrevlett.79.325).
- [16] A. Abbas et al., "Challenges and opportunities in quantum optimization," 2023, *arXiv:2312.02279*.
- [17] C. Mastroianni, F. Plastina, L. Scarcello, J. Settino, and A. Vinci, "Assessing quantum computing performance for energy optimization in a prosumer community," *IEEE Trans. Smart Grid*, vol. 15, no. 1, pp. 444–456, Jan. 2024.

- [18] M. Federer, D. Müssig, S. Klaiber, J. Lässig, P. Bretschneider, and S. Lenk, "Application benchmark for quantum optimization on electromobility use case," in *Proc. IEEE Vehicle Power Propuls. Conf. (VPPC)*, Nov. 2022, pp. 1–6.
- [19] E. Farhi, J. Goldstone, and S. Gutmann, "A quantum approximate optimization algorithm," 2014, *arXiv:1411.4028*.
- [20] L. Zhou, S.-T. Wang, S. Choi, H. Pichler, and M. D. Lukin, "Quantum approximate optimization algorithm: Performance, mechanism, and implementation on near-term devices," *Phys. Rev. X*, vol. 10, no. 2, Jun. 2020, Art. no. 021067, doi: [10.1103/physrevx.10.021067](https://doi.org/10.1103/physrevx.10.021067).
- [21] J. Preskill, "Quantum computing in the NISQ era and beyond," *Quantum*, vol. 2, p. 79, Aug. 2018. [Online]. Available: <https://quantum-journal.org/papers/q-2018-08-06-79/>
- [22] A. Lucas, "Ising formulations of many NP problems," *Frontiers Phys.*, vol. 2, pp. 1–14, Jan. 2014. [Online]. Available: <https://www.frontiersin.org/journals/physics/articles/10.3389/fphy.2014.00005>
- [23] J. R. McClean, J. Romero, R. Babbush, and A. Aspuru-Guzik, "The theory of variational hybrid quantum-classical algorithms," *New J. Phys.*, vol. 18, no. 2, Feb. 2016, Art. no. 023023. [Online]. Available: <https://iopscience.iop.org/article/10.1088/1367-2630/18/2/023023>
- [24] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles, "Variational quantum algorithms," *Nature Rev. Phys.*, vol. 3, no. 9, pp. 625–644, Aug. 2021, doi: [10.1038/s42254-021-00348-9](https://doi.org/10.1038/s42254-021-00348-9).
- [25] K. Bharti, A. Cervera-Lierta, T. H. Kyaw, T. Haug, S. Alperin-Lea, A. Anand, M. Degroote, H. Heimonen, J. S. Kottmann, T. Menke, W.-K. Mok, S. Sim, L.-C. Kwek, and A. Aspuru-Guzik, "Noisy intermediate-scale quantum algorithms," *Rev. Modern Phys.*, vol. 94, no. 1, pp. 1–16, Feb. 2022, doi: [10.1103/revmodphys.94.015004](https://doi.org/10.1103/revmodphys.94.015004).
- [26] R. H. Byrd, P. Lu, J. Nocedal, and C. Zhu, "A limited memory algorithm for bound constrained optimization," *SIAM J. Sci. Comput.*, vol. 16, no. 5, pp. 1190–1208, Sep. 1995, doi: [10.1137/0916069](https://doi.org/10.1137/0916069).
- [27] H. Singh, S. Majumder, and S. Mishra, "Benchmarking of different optimizers in the variational quantum algorithms for applications in quantum chemistry," *J. Chem. Phys.*, vol. 159, no. 4, Jul. 2023, Art. no. 044117. [Online]. Available: <https://pubs.aip.org/jcp/article/159/4/044117/2904876/Benchmarking-of-different-optimizers-in-the>
- [28] P. Díez-Valle, D. Porras, and J. J. García-Ripoll, "Quantum variational optimization: The role of entanglement and problem hardness," *Phys. Rev. A, Gen. Phys.*, vol. 104, no. 6, pp. 1–18, Dec. 2021, doi: [10.1103/physreva.104.062426](https://doi.org/10.1103/physreva.104.062426).
- [29] W. Lavrijsen, A. Tudor, J. Müller, C. Iancu, and W. de Jong, "Classical optimizers for noisy intermediate-scale quantum devices," in *Proc. IEEE Int. Conf. Quantum Comput. Eng. (QCE)*, Denver, CO, USA, Oct. 2020, pp. 267–277. [Online]. Available: <https://ieeexplore.ieee.org/document/9259985/>
- [30] J. Romero, R. Babbush, J. R. McClean, C. Hempel, P. J. Love, and A. Aspuru-Guzik, "Strategies for quantum computing molecular energies using the unitary coupled cluster ansatz," *Quantum Sci. Technol.*, vol. 4, no. 1, Oct. 2018, Art. no. 014008, doi: [10.1088/2058-9565/aad3e4](https://doi.org/10.1088/2058-9565/aad3e4).
- [31] J. Stokes, J. Isaac, N. Killoran, and G. Carleo, "Quantum natural gradient," *Quantum*, vol. 4, p. 269, May 2020, doi: [10.22331/q-2020-05-25-269](https://doi.org/10.22331/q-2020-05-25-269).
- [32] D. Faifde, J. D. Viqueira, M. M. Juane, and A. Gómez, "Using differential evolution to avoid local minima in variational quantum algorithms," *Sci. Rep.*, vol. 13, no. 1, p. 16230, Sep. 2023, doi: [10.1038/s41598-023-43404-3](https://doi.org/10.1038/s41598-023-43404-3).
- [33] M. Powell, "A direct search optimization method that models the objective and constraint functions by linear interpolation," in *Advances in Optimization and Numerical Analysis*. The Netherlands: Springer, 1994, pp. 51–67.
- [34] R. Storn and K. Price, "Differential evolution—A simple and efficient heuristic for global optimization over continuous spaces," *J. Global Optim.*, vol. 11, pp. 341–359, Jan. 1997.
- [35] A. Peruzzo, J. McClean, P. Shadbolt, M.-H. Yung, X.-Q. Zhou, P. J. Love, A. Aspuru-Guzik, and J. L. O'Brien, "A variational eigenvalue solver on a photonic quantum processor," *Nature Commun.*, vol. 5, no. 1, pp. 1–23, Jul. 2014, doi: [10.1038/ncomms5213](https://doi.org/10.1038/ncomms5213).
- [36] G. Nannicini, "Performance of hybrid quantum-classical variational heuristics for combinatorial optimization," *Phys. Rev. E, Stat. Phys. Plasmas Fluids Relat. Interdiscip. Top.*, vol. 99, no. 1, Jan. 2019, doi: [10.1103/physreve.99.013304](https://doi.org/10.1103/physreve.99.013304).
- [37] E. Farhi and A. W. Harrow, "Quantum supremacy through the quantum approximate optimization algorithm," 2016, *arXiv:1602.07674*.
- [38] G. Verdon, M. Broughton, and J. Biamonte, "A quantum algorithm to train neural networks using low-depth circuits," 2017, *arXiv:1712.05304*.
- [39] P. Díez-Valle, D. Porras, and J. J. García-Ripoll, "Quantum approximate optimization algorithm pseudo-Boltzmann states," *Phys. Rev. Lett.*, vol. 130, no. 5, Feb. 2023, Art. no. 050601, doi: [10.1103/physrevlett.130.050601](https://doi.org/10.1103/physrevlett.130.050601).
- [40] E. Farhi, J. Goldstone, S. Gutmann, and L. Zhou, "The quantum approximate optimization algorithm and the sherrington-kirkpatrick model at infinite size," *Quantum*, vol. 6, p. 759, Jul. 2022. [Online]. Available: <https://quantum-journal.org/papers/q-2022-07-07-759/>
- [41] J. Claes and W. V. Dam, "Instance independence of single layer quantum approximate optimization algorithm on mixed-spin models at infinite size," *Quantum*, vol. 5, p. 542, Sep. 2021. [Online]. Available: <https://quantum-journal.org/papers/q-2021-09-15-542/>
- [42] V. Akshay, D. Rabinovich, E. Campos, and J. Biamonte, "Parameter concentrations in quantum approximate optimization," *Phys. Rev. A, Gen. Phys.*, vol. 104, no. 1, Jul. 2021, Art. no. 010401, doi: [10.1103/physreva.104.010401](https://doi.org/10.1103/physreva.104.010401).
- [43] M. Streif and M. Leib, "Forbidden subspaces for level-1 quantum approximate optimization algorithm and instantaneous quantum polynomial circuits," *Phys. Rev. A, Gen. Phys.*, vol. 102, no. 4, Oct. 2020, Art. no. 042416, doi: [10.1103/physreva.102.042416](https://doi.org/10.1103/physreva.102.042416).
- [44] G. E. Crooks, "Performance of the quantum approximate optimization algorithm on the maximum cut problem," 2018, *arXiv:1811.08419*.
- [45] M. P. Harrigan et al., "Quantum approximate optimization of non-planar graph problems on a planar superconducting processor," *Nature Phys.*, vol. 17, no. 3, pp. 332–336, Feb. 2021. [Online]. Available: <https://www.nature.com/articles/s41567-020-01105-y>
- [46] E. Pelofske, A. Bärttschi, J. Golden, and S. Eidenbenz, "High-round QAOA for MAX k -SAT on trapped ion NISQ devices," in *Proc. IEEE Int. Conf. Quantum Comput. Eng. (QCE)*, Bellevue, WA, USA, Sep. 2023, pp. 506–517. [Online]. Available: <https://ieeexplore.ieee.org/document/10313612/>
- [47] S. Ebadi et al., "Quantum optimization of maximum independent set using Rydberg atom arrays," *Science*, vol. 376, no. 6598, pp. 1209–1215, Jun. 2022. [Online]. Available: <https://www.science.org/doi/10.1126/science.abb6587>
- [48] P. Niroula, R. Shaydulin, R. Yalovetzky, P. Minssen, D. Herman, S. Hu, and M. Pistoia, "Constrained quantum optimization for extractive summarization on a trapped-ion quantum computer," *Sci. Rep.*, vol. 12, no. 1, p. 17171, Oct. 2022. [Online]. Available: <https://www.nature.com/articles/s41598-022-20853-w>
- [49] J. S. Baker and S. K. Radha, "Wasserstein solution quality and the quantum approximate optimization algorithm: A portfolio optimization case study," 2022, *arXiv:2202.06782*.
- [50] D. Amaro, M. Rosenkranz, N. Fitzpatrick, K. Hirano, and M. Fiorentini, "A case study of variational quantum algorithms for a job shop scheduling problem," *EPJ Quantum Technol.*, vol. 9, no. 1, pp. 1–24, Feb. 2022, doi: [10.1140/epjqt/s40507-022-00123-4](https://doi.org/10.1140/epjqt/s40507-022-00123-4).
- [51] T. Lubinski, C. Coffrin, C. McGeoch, P. Sathe, J. Apanavicius, D. Bernal Neira, and Q. Economic Development Consortium(QED-C) Collaboration, "Optimization applications as quantum performance benchmarks," *ACM Trans. Quantum Comput.*, vol. 5, no. 3, pp. 1–44, Aug. 2024, doi: [10.1145/3678184](https://doi.org/10.1145/3678184).
- [52] P. K. Barkoutsos, G. Nannicini, A. Robert, I. Tavernelli, and S. Woerner, "Improving variational quantum optimization using CVaR," *Quantum*, vol. 4, p. 256, Apr. 2020, doi: [10.22331/q-2020-04-20-256](https://doi.org/10.22331/q-2020-04-20-256).
- [53] L. Li, M. Fan, M. Coram, P. Riley, and S. Leichenauer, "Quantum optimization with a novel Gibbs objective function and ansatz architecture search," *Phys. Rev. Res.*, vol. 2, no. 2, Apr. 2020, Art. no. 023074, doi: [10.1103/physrevresearch.2.023074](https://doi.org/10.1103/physrevresearch.2.023074).
- [54] R. Diestel, *Extremal Graph Theory*. Berlin, Germany: Springer, 2025, pp. 179–226, doi: [10.1007/978-3-662-70107-2_7](https://doi.org/10.1007/978-3-662-70107-2_7).
- [55] G. Scriva, N. Astrakhantsev, S. Pilati, and G. Mazzola, "Challenges of variational quantum optimization with measurement shot noise," *Phys. Rev. A, Gen. Phys.*, vol. 109, no. 3, Mar. 2024, Art. no. 032408, doi: [10.1103/physreva.109.032408](https://doi.org/10.1103/physreva.109.032408).
- [56] J. F. Gonthier, M. D. Radin, C. Buda, E. J. Doskocil, C. M. Abuan, and J. Romero, "Measurements as a roadblock to near-term practical quantum advantage in chemistry: Resource analysis," *Phys. Rev. Res.*, vol. 4, no. 3, Aug. 2022, Art. no. 033154, doi: [10.1103/physrevresearch.4.033154](https://doi.org/10.1103/physrevresearch.4.033154).

- [57] A. Ozaeta, W. van Dam, and P. L. McMahon, "Expectation values from the single-layer quantum approximate optimization algorithm on ising problems," *Quantum Sci. Technol.*, vol. 7, no. 4, Sep. 2022, Art. no. 045036, doi: [10.1088/2058-9565/ac9013](https://doi.org/10.1088/2058-9565/ac9013).
- [58] P. Díez-Valle, D. Porras, and J. J. García-Ripoll, "Connection between single-layer quantum approximate optimization algorithm interferometry and thermal distribution sampling," *Frontiers Quantum Sci. Technol.*, vol. 3, pp. 1–24, Feb. 2024, doi: [10.3389/frqst.2024.1321264](https://doi.org/10.3389/frqst.2024.1321264).
- [59] A. Javadi-Abhari, M. Treinish, K. Krsulich, C. J. Wood, J. Lishman, J. Gacon, S. Martiel, P. D. Nation, L. S. Bishop, A. W. Cross, B. R. Johnson, and J. M. Gambetta, "Quantum computing with qiskit," 2024, *arXiv:2405.08810*.
- [60] E. Farhi, D. Gamarnik, and S. Gutmann, "The quantum approximate optimization algorithm needs to see the whole graph: Worst case examples," 2020, *arXiv:2005.08747*.
- [61] R. Herrman, P. C. Lotshaw, J. Ostrowski, T. S. Humble, and G. Siopsis, "Multi-angle quantum approximate optimization algorithm," *Sci. Rep.*, vol. 12, no. 1, p. 6781, Apr. 2022. [Online]. Available: <https://www.nature.com/articles/s41598-022-10555-8>
- [62] S. Bravyi, A. Kliesch, R. Koenig, and E. Tang, "Obstacles to variational quantum optimization from symmetry protection," *Phys. Rev. Lett.*, vol. 125, no. 26, Dec. 2020, Art. no. 260505, doi: [10.1103/physrevlett.125.260505](https://doi.org/10.1103/physrevlett.125.260505).
- [63] M. Sciorilli, L. Borges, T. L. Patti, D. García-Martín, G. Camilo, A. Anandkumar, and L. Aolita, "Towards large-scale quantum optimization solvers with few qubits," 2024, *arXiv:2401.09421*.
- [64] B. Tan, M.-A. Lemonde, S. Thanasilp, J. Tangpanitanon, and D. G. Angelakis, "Qubit-efficient encoding schemes for binary optimisation problems," *Quantum*, vol. 5, p. 454, May 2021. [Online]. Available: <https://quantum-journal.org/papers/q-2021-05-04-454/>
- [65] E. X. Huber, B. Y. L. Tan, P. R. Griffin, and D. G. Angelakis, "Exponential qubit reduction in optimization for financial transaction settlement," 2023, *arXiv:2307.07193*.
- [66] I. D. Leonidas, A. Dukakis, B. Tan, and D. G. Angelakis, "Qubit efficient quantum algorithms for the vehicle routing problem on noisy intermediate-scale quantum processors," *Adv. Quantum Technol.*, vol. 7, no. 5, May 2024, Art. no. 2300309. [Online]. Available: <https://onlinelibrary.wiley.com/doi/10.1002/quote.202300309>
- [67] Y. Tene-Cohen, T. Kelman, O. Lev, and A. Makmal, "A variational qubit-efficient MaxCut heuristic algorithm," 2023, *arXiv:2308.10383*.
- [68] C. Gambella and A. Simonetto, "Multiblock ADMM heuristics for mixed-binary optimization on classical and quantum computers," *IEEE Trans. Quantum Eng.*, vol. 1, pp. 1–22, 2020.
- [69] T. Viet Le and V. Kekatos, "Variational quantum eigensolver with constraints (VQEC): Solving constrained optimization problems via VQE," 2023, *arXiv:2311.08502*.
- [70] D. Herman, R. Shaydulin, Y. Sun, S. Chakrabarti, S. Hu, P. Minssen, A. Rattew, R. Yalovetzky, and M. Pistoia, "Constrained optimization via quantum Zeno dynamics," *Commun. Phys.*, vol. 6, no. 1, p. 219, Aug. 2023. [Online]. Available: <https://www.nature.com/articles/s42005-023-01331-9>
- [71] T. Hao, R. Shaydulin, M. Pistoia, and J. Larson, "Exploiting in-constraint energy in constrained variational quantum optimization," in *Proc. IEEE/ACM 3rd Int. Workshop Quantum Comput. Softw. (QCS)*, Nov. 2022, pp. 100–106, doi: [10.1109/QCS56647.2022.00017](https://doi.org/10.1109/QCS56647.2022.00017).
- [72] P. Díez-Valle, J. Luis-Hita, S. Hernández-Santana, F. Martínez-García, Á. Díaz-Fernández, E. Andrés, J. José García-Ripoll, E. Sánchez-Martínez, and D. Porras, "Multiobjective variational quantum optimization for constrained problems: An application to cash handling," *Quantum Sci. Technol.*, vol. 8, no. 4, Jul. 2023, Art. no. 045009, doi: [10.1088/2058-9565/ace474](https://doi.org/10.1088/2058-9565/ace474).

MATEO ALONSO received the B.S. degree in physics from the University of Santiago de Compostela, in 2019, and the M.S. degree in physics engineering from the Universitat Politècnica de Catalunya, in 2021. He is currently pursuing the Ph.D. degree in quantum computing. In addition, he has complementary training in machine and deep learning, SQL for data science, and quantum computing. He was a Junior Researcher with CITMaga, the Galician Center for Mathematical Research and Technology, from 2021 to 2022, when he joined ITG as a Data and Quantum Scientist.

GUILLERMO RUBIÑOS RODRÍGUEZ received the B.S. degree in physics from the Universidade de Santiago de Compostela, in 2024. He is currently pursuing the M.S. degree in quantum technologies with CSIC. He is a Junior Researcher with the Data and Quantum Team, ITG. He has complementary training in quantum computing and quantum information systems.

PABLO DÍEZ-VALLE received the B.S. degree in physics and the M.S. degree in theoretical and mathematical physics. He is currently pursuing the Ph.D. degree in quantum computing algorithms applied to combinatorial optimization and generative modeling problems with CSIC. His academic studies reinforces his strong mathematical background and problem-solving skills. He joined ITG's Quantum Team, in 2024. He is the first author of seven scientific papers in the field of quantum algorithms.

ANA GARBAYO received the bachelor's degree in physics from the Universidade de Santiago de Compostela (USC), in 2017, the master's degree in high energy physics from l'École Polytechnique of Paris, in 2018, and the Ph.D. degree in theoretical physics from USC, in 2022. After completing her doctorate, she was with the consulting sector, focusing on energy-related business environments, from 2022 to 2024. In 2024, she joined ITG as a Data and Quantum Scientist. She has published three articles in high-impact journals indexed in the Journal Citation Reports (JCR) and participates in several vocational, educational, and teaching activities to promote scientific careers.

XELA GARCÍA-SANTIAGO received the B.S. degree in chemical engineering, the two M.S. degrees in renewable energy and sustainability and in industrial mathematics, and the Ph.D. degree in environmental and chemical engineering from the Universidade de Santiago de Compostela (USC), in 2010, 2015, and 2019, respectively. During her career, she has combined research and industry positions, being a Senior Research Support Technician at the university, a Production Engineer with Ferroalntica, and a Researcher with the USC Environmental Biotechnology Group. In 2021, she joined the AI & ICT Division, ITG, as a Data Engineer. She is the author of eight scientific articles and has participated in several research projects, including HE and LIFE programs.

GONZALO BLÁZQUEZ GIL received the B.S. and M.S. degrees from the Universidad Carlos III de Madrid, and the Ph.D. degree in computer science from the Universidad Carlos III de Madrid, in 2015. He is a Computer Engineer. From 2007 to 2013, he was a Research Assistant with the Applied Artificial Intelligence Group. He is the Technical Leader of the Data Science and Quantum Technologies Research Team. Currently, he is the AI and Operations Manager with ITG and the Scientific Advisor of FlyThings, ITG's IoT spin-off. Since 2023, he is an Associate Professor with the Universidad Europea de Madrid. Since 2014, he has been a Researcher with Galicia Technology Institute, Artificial Intelligence and Information Systems Division. He has been involved in more than 40 research, development, and innovation projects. He is the author of 18 papers and conference communications, focusing on industrial IoT, big data, cyber-physics systems, AI forecasting models and quantum computing, and cryptography. His main fields of specialization were activity and emotion recognition, context-based systems, and machine learning and computer vision.

...